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Simultaneous quantitative analysis of multiple metabolites using label-free surface-enhanced Raman spectroscopy and explainable deep learning

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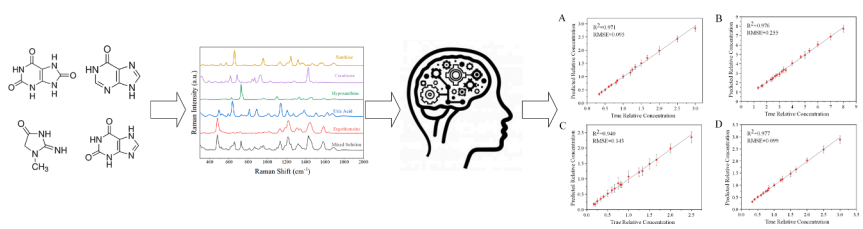
HIGHLIGHTS

- Successful simultaneous quantitative detection of multiple metabolites, maintaining potential efficacy with an increased number of targets.
- An explainable DNN model using multi-channel feature extraction identified distinct peaks for each target metabolite from mixture spectra.
- SERS spectra of a multi-metabolite mixture were successfully obtained using a single-layer silver nanoparticle substrate.

GRAPHICAL ABSTRACT



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ABSTRACT

Metabolites serve as vital biomarkers, reflecting physiological and pathological states and offering insights into disease progression and early detection. This study introduces an advanced analytical technique integrating label-free Surface-Enhanced Raman Spectroscopy (SERS) with deep learning, and leverages SHAP (SHapley Additive exPlanations) to provide a visual interpretative analysis of the predictive rationale of the deep learning model, facilitating simultaneous detection and quantitative analysis of multiple metabolites. Monolayer silver nanoparticle SERS substrates were fabricated via a triple-phase interfacial self-assembly method, which captured complex spectral information of target metabolites in mixed solutions. A custom-built deep neural network model with multi-channel feature extraction was employed to predict the concentrations of uric acid (R^2 = 0.976), xanthine (R^2 = 0.971), hypoxanthine (R^2 = 0.977), and creatinine (R^2 = 0.940). The method's scalability was validated as the performance remained consistent with an increasing number of simultaneous targets. This approach offers a sensitive, cost-effective, and rapid alternative for metabolite analysis, with significant implications for clinical diagnostics and personalized medicine.

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1. Introduction

Metabolites are pivotal biomarkers that reflect physiological and pathological states, offering insights into the early identification of disease markers and the progression of illnesses [1]. Metabolomics, as a pivotal discipline at the intersection of systems biology, plays an essential role in elucidating the complex interplay between cellular processes and metabolic pathways. By harnessing extensive biological information from both patient cohorts and healthy populations, this field has catalyzed remarkable advancements in the discovery and characterization of biomarkers, with a particular emphasis on the early detection and classification of cancer. These strides underscore the transformative potential of metabolomics in advancing our understanding of cancer biology and facilitating the development of novel diagnostic and therapeutic strategies [2]. Recent studies have highlighted that metabolites released by cancerous cells and those from patients with breast, prostate, pancreatic, and lung cancers differ remarkably from those of healthy individuals, underscoring the potential of metabolite-based diagnostics for cancer [3–7]. Metabolite detection thus holds considerable promise for both fundamental research and clinical applications [8].

Current mainstream techniques for metabolite separation and detection in biological fluids include mass spectrometry, nuclear magnetic resonance, and their hyphenated methods with chromatography [9,10]. However, these approaches demand considerable costs and time for complex sample processing and analysis, severely restricting their clinical translation and application. There is an urgent need for a metabolite detection method that is cost-effective, straightforward, rapid, highly sensitive, and capable of multiplex detection for continuous, real-time monitoring.

Surface-Enhanced Raman Spectroscopy (SERS) emerges as an ultrasensitive and specific optical detection technique, capable of amplifying Raman signals by 10^9 to 10^{15} times in the vicinity of nanostructures through localized surface plasmon resonance [11,12]. This technology enables chemical fingerprint analysis via molecular vibration signals and is characterized by its sensitivity, label-free nature, speed, and specificity [13]. SERS spectra from complex systems like body fluids can reveal key molecular information on the relative abundance of components, facilitating the monitoring of changes in relative composition within mixtures. SERS-based detection techniques can be categorized into direct (label-free) and indirect (label, with SERS tags) methods [14]. The indirect strategy involves using nanoparticles modified with Raman reporter molecules as tags for specific binding to targets, such as protein biomarkers. Here, target detection relies on the enhanced signal from the Raman reporters on the nanoparticles. In spite of the widespread application of labeled SERS for detecting macromolecules and cells [15,16], metabolite detection typically employs a label-free direct approach [17]. Label-free SERS has shown immense potential in cancer detection and screening [18], with reports on its application in classifying cancers such as breast, prostate, and bladder [19–21]. However, due to challenges in quantification, these methods have been primarily qualitative, and they have not yet enabled the quantitative differentiation or typing of cancer based on changes in metabolite content from spectra.

Most research on SERS-based metabolite quantification concentrates on the straightforward quantification of single target substances within a simplified system. Improvements in substrate materials and structures enhance the enhancement factor and reduce the contribution of non-target metabolites to mixed spectra, establishing a linear relationship between specific spectral features and concentration, thus providing a basis for quantification [22]. Machine learning and deep learning serve as powerful tools in spectral analysis, offering a more rapid and accurate means of quantification than manual fitting of linear relationships between spectral peak intensity and concentration [17]. The combination of SERS with machine learning, such as Partial Least Squares (PLS) regression, or deep learning for quantitative detection, has been widely reported in food [23–25], environmental, and pharmaceutical detection [26,27]. However, these methods have primarily been applied to the quantification of specific target substances in simple solution systems. The use of label-free SERS substrates to measure complex spectra with multiple components presents significant challenges due to spectral overlap and varying affinities of different substances for the SERS substrate, complicating quantitative analysis. In the presence of complex systems with overlapping characteristic peaks, most research is concentrated on classification problem, which is qualitative analysis [28]. In 2021, an analysis of the SERS spectra of urine from pancreatic and prostate cancer patients revealed differences primarily due to variations in the intensity of SERS peaks of six metabolites, including uric acid, hypoxanthine, and xanthine. Unsupervised principal component analysis and supervised orthogonal partial least squares discriminant analysis were employed for multivariate analysis of SERS spectra, successfully distinguishing cancer patients from healthy individuals [29]. In 2024, the latest reports have demonstrated the ability to discriminate the presence of three antibiotics—ciprofloxacin, doxycycline, and levofloxacin—in mixed drugs and to quantitatively determine their ratios under ten simple mixing conditions using Convolutional Neural Networks (CNN) and Neural Network-Expectation Maximization (NN-EM) models [30]. However, there are limited reports on the independent analysis of multiple spectral components within complex SERS spectra to achieve simultaneous quantification of the components. There is a clear necessity to develop simultaneous quantitative detection techniques for multiple metabolites within complex systems.

This study introduces a novel method for the simultaneous quantitative analysis of multiple metabolites based on the integration of label-free SERS technology and deep learning for spectral data analysis. We employed a three-phase interface self-assembly technique to fabricate a single-layered silver nanoparticle film as a SERS substrate on single-crystal silicon. This substrate successfully captured complex SERS spectra containing spectral information of all target metabolites in a mixed solution, including uric acid, xanthine, hypoxanthine, and creatinine, which are closely associated with pancreatic, prostate, and renal cancers, with ergothioneine added as an additional spectral interferent. We constructed a neural network model incorporating a multi-channel extraction mechanism, which is distinct from common neural networks, and combined it with the internal standard method [31,32]. Employing the SHAP method, the internal rationale for concentration predictions by the model was visualized, demonstrating the model's successful extraction of characteristic peaks corresponding to each quantitative target metabolite within the mixture spectrum, and achieving simultaneous quantitative prediction of four metabolites through information such as the intensity of the corresponding characteristic peaks. The determination coefficients (R^2) for the predicted concentrations versus their actual mixed concentrations were as follows: xanthine: 0.971; hypoxanthine: 0.977; uric acid: 0.976; and creatinine: 0.941. Importantly, as the number of targets for simultaneous quantitative detection increased, the quantitative performance for each individual target did not significantly deteriorate, indicating the scalability of this method for the inclusion of additional metabolites in future studies. Our approach presents a promising strategy for the simultaneous quantitative detection of multiple metabolites using label-free SERS technology.

2. Experimental section

2.1. Fabrication of Ag-NPs SERS substrates

Ag-NPs were prepared with reference to previous work [33]. Preparation of Ag NPs SERS substrates by three-phase interfacial self-assembly method [34].

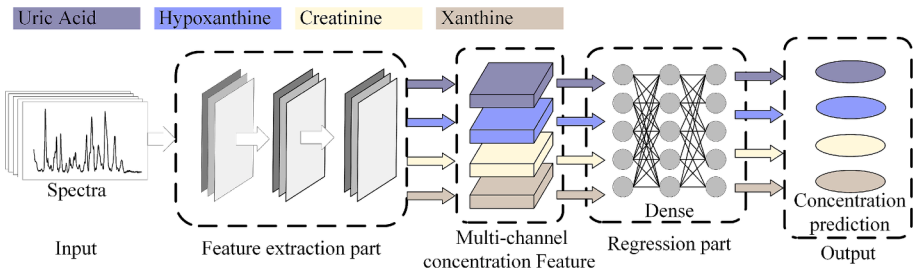

Fig. 1. Structure of the multi-channel feature extraction neural network model.

2.2. SERS determination of mixed solution

Solid samples of hypoxanthine, creatinine, sodium xanthate, sodium urate, and ergothioneine were accurately weighed and dissolved in Phosphate-Buffered Saline (PBS) to prepare high-concentration stock solutions. The solutions were subjected to ultrasonication to ensure complete dissolution and then stored at 4 °C for subsequent use. The sources of reagents and materials used in the experiments are detailed in the Supplementary materials. Each metabolite was prepared within predefined concentration ranges: hypoxanthine (5–15 μM), creatinine (250–1250 μM), xanthine (25–75 μM), and uric acid (80–160 μM). For each metabolite, the concentrations were divided into five distinct gradient levels. The concentration levels for each metabolite were then randomly shuffled and combined to create different concentration schemes. These schemes were numbered and served as the basis for generating concentration labels used in subsequent neural network training. The data in the concentration table represent the volumes of the high-concentration stock solutions that need to be added to 5 ml centrifuge tubes to prepare the complex mixed solutions. According to the predefined concentration schemes, the appropriate amounts of each high-concentration stock solution were calculated and added to the centrifuge tubes. Ergothioneine was randomly added to the mixtures. Finally, the mixtures were adjusted to a final volume of 4 ml with pure PBS to create the final complex mixed solutions.

The SERS substrates were immersed in the mixed solution for 2 min, ensuring thorough interaction. After retrieval, the substrates were rinsed with PBS to eliminate any residual solution. Subsequently, they were briefly immersed in ethanol, which allowed for rapid drying due to its volatile nature, facilitating the immediate collection of the SERS spectra.

The Renishaw inVia confocal Raman spectrometer was utilized for spectral detection. Calibration of the instrument was performed using a silicon with peak at 520 cm^{-1} prior to each measurement session. The SERS spectra were recorded under standardized conditions: a 20 × objective lens; a laser wavelength of 785 nm; a laser power of 4.8 mW; an exposure time of 10 s; and an accumulation of three scans per spectrum to enhance the signal-to-noise ratio (SNR). The spectral range of 300–2000 cm^{-1} was targeted for data collection, which was then utilized for analysis and quantitative modeling.

2.3. Model architecture and training strategy

This section delineates the methodological framework of our study, encompassing three critical stages: the preprocessing and assembly of a quantitative spectral dataset, the construction and fine-tuning of a deep neural network architecture, and the training and validation of the model.

Our approach to spectral data preprocessing began with the utilization of WIRE4.3 software, an integral component of our spectrometer, to systematically address noise reduction and cosmic ray removal from the acquired spectral data. Subsequent to this initial data cleansing, an Asymmetric Least Square (ALS) baseline correction technique was applied to refine the spectral profiles. The resultant spectra underwent a series of normalization and standardization processes, culminating in the final curated dataset.

For the precise assignment of concentration labels, we employed the internal standard method. The concentration labels for xanthine, uric acid, and creatinine were calculated as the ratio of their respective concentrations to that of hypoxanthine. Conversely, the concentration label for hypoxanthine was determined based on the ratio of its concentration to that of xanthine. This method, leveraging relative spectral intensities to enhance substance quantification, is well-supported by the existing literature and has proven effective in analogous contexts [31,32].

To fulfill the requirement for simultaneous quantitative analysis of multiple targets, we developed a multi-channel convolutional neural network (CNN) tailored for regression tasks. The operational flow of the model is diagrammatically divided into four principal stages: spectral data input, multi-channel spectral feature extraction, regression analysis of spectral features, and output of concentration prediction results.

In the feature extraction phase, the model employs multiple one-dimensional convolutional layers to extract features from the input spectral data. These layers perform feature extraction according to the number of quantifiable targets, segregating outputs across distinct channels for each. The convolutional layers use tunable hyperparameters specifically, the number of filters ranging between 32 and 512, and kernel sizes adjustable between 1 × 1 and 5 × 5. Each convolutional layer is followed by a non-linear activation function, with options including ReLU and linear activation. These hyperparameters are optimized using Bayesian optimization with the Hyperband tuner, allowing the model to capture features at multiple levels of granularity and automatically discover the most relevant features for the task.

The extracted features are subsequently fed into an information mapping module, comprised of fully connected layers that execute regression computations on the concentration data. The model utilizes a custom coefficient of determination (R^2) loss function to ensure that the predictions not only minimize the mean squared error but also maintain the variability of the target data.

The model was trained using the Adam optimizer. The Bayesian optimization process systematically evaluates different combinations of hyperparameters such as the number of filters, kernel size, and activation functions and selects the configuration that achieves the best validation performance. This approach ensures that the model is highly tuned to the dataset, improving its generalization ability on unseen data (see Fig. 1).

The efficacy of our model was assessed through a rigorous 10-fold cross-validation methodology, partitioning the spectral dataset into distinct training and testing subsets. This process yielded a robust estimate of the model's predictive capabilities within the complex spectral landscape. The predictive prowess of the model is gauged by two metrics: the coefficient of determination (R^2), measuring the correlation between predicted and actual concentrations of each substance, and the root mean square error (RMSE), quantifying the precision of the model's predictions. In the pursuit of an interpretable analysis of model predictions, we have utilized the SHAP (SHapley Additive exPlanations) methodology to graphically elucidate the underpinnings of the model's predictive decisions. SHAP is a game theoretic approach to explain the

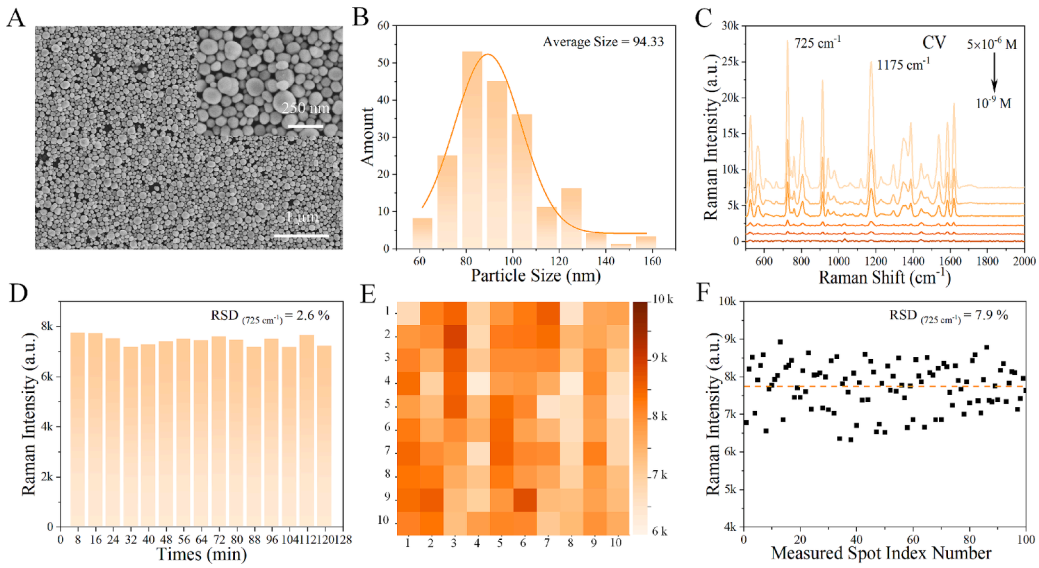

Fig. 2. Characterization and performance evaluation of the Ag NPs SERS substrate. (A) The SEM image of Ag nanoparticles. (B) The size statistics of Ag nanoparticles (C) SERS spectra of crystal violet (CV) molecules at concentrations ranging from 5×10^{-6} M to 10^{-9} M on the Ag NPs SERS substrate. (D) Time stability assessment of the Ag NPs SERS substrate under continuous low-power laser irradiation (30 mW). (E and F) SERS intensity mapping of the CV molecule Raman characteristic peak (725 cm^{-1}) across 100 random points on the Ag NPs SERS substrate.

output of Deep Learning model. It connects optimal credit allocation with local explanations using the classic Shapley values from game theory and their related extensions [35]. Collectively, these metrics provide a comprehensive evaluation of the model's performance in the domain of spectral analysis.

3. Results and discussion

3.1. SERS substrate characterization

We employed the method of reducing silver nitrate using trisodium citrate and ascorbic acid to prepare Ag nanoparticles (NPs). The prepared Ag NPs were assembled into a monolayer SERS substrate using a three-interface self-assembly method, as shown in Fig. 2A. It can be clearly observed that the Ag NPs exhibit irregular spherical shapes with tight packing between them, with an average size of 94.33 nm. Utilizing crystal violet (CV) as the probe molecule, the performance of the Ag NPs SERS substrate was investigated. The SERS spectra of CV molecules ranging from 5×10^{-6} M to 10^{-9} M on the Ag NPs SERS substrate were shown in Fig. 2C. The results indicate that the detection limit of the Ag NPs SERS substrate for CV molecules is as low as 10^{-9} M. Additionally, the time stability of the Ag NPs SERS substrate was illustrated in Fig. 2D. The Ag NPs SERS substrate was fixed under the laser probe, and spectra were collected every 8 min. The results demonstrate that under continuous irradiation of a low-power laser (30 mW), the Ag NPs SERS substrate maintains good stability. Similarly, the uniformity of the SERS substrate is also an important criterion. We randomly collected SERS spectra from 100 points on the Ag NPs SERS substrate and plotted mapping of the intensity of the CV molecule Raman characteristic peak (725 cm^{-1}) (Fig. 2E and F). The results indicate that SERS substrates prepared using the interface self-assembly method exhibit excellent uniformity, as has been confirmed in numerous studies. This provides the basis for the subsequent acquisition of large and stable spectral data.

The batch-to-batch reproducibility and storage stability of SERS substrates is particularly important for quantitative detection. Therefore, SERS spectra of different batches and storage days of SERS substrates were displayed in Fig.S2 (Supporting Information). Due to the superiority of the assembly method, the nine batches of SERS substrates were prepared completely and the SERS homogeneity was excellent with an RSD value of 9.38 %. In addition, this SERS substrate has excellent storage stability, providing stable SERS activity over 7 days of storage (Fig. S2, Supporting Information).

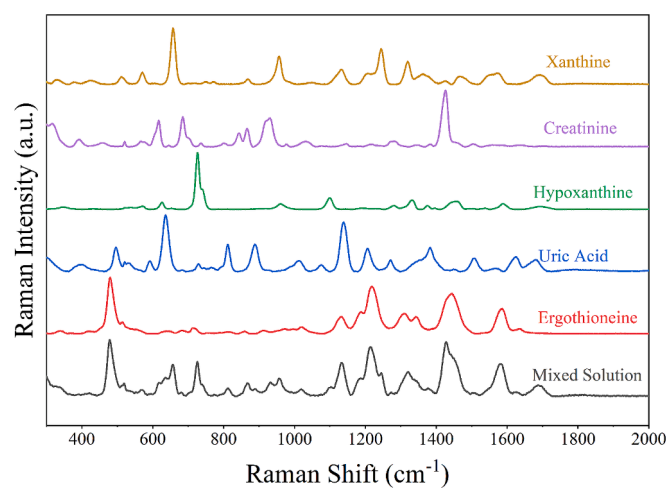

Fig. 3. Spectra of pure metabolites and mixture solutions.

3.2. Determination of target metabolites using Ag NPs Sensor

During the experimental process, it was observed that the SERS spectra of ergothioneine and hypoxanthine underwent changes over time following their extraction. As time progressed, the characteristic peak of xanthine at 725 cm^{-1} gradually diminished in intensity, while the adjacent peak at 738 cm^{-1} intensified. Conversely, for ergothioneine, the characteristic peaks at 478 cm^{-1} and 1433 cm^{-1} decreased, while the peak at 1312 cm^{-1} became more pronounced, eventually becoming the most prominent feature in the SERS spectrum. The overall spectral shape of ergothioneine transformed into a new spectral pattern, as detailed in Supplementary Material Fig. S1. To ensure that these variations do not impact quantification, SERS spectra collection must be conducted immediately after the substrates are dried post-immersion.

We collected pure substance spectra of each metabolite in PBS buffer

Table 1
Model performance for single-target prediction.

Metabolites	Xanthine	Hypoxanthine	Uric Acid	Creatinine
R^2	0.972	0.973	0.971	0.938
RMSE	0.097	0.096	0.251	0.152

Table 2
Incremental quantitative prediction performance of multi-channel neural network models for target metabolites.

Simultaneous Prediction of Target Quantities	2	3	4
Predictive Performance	uric acid R^2 0.973 uric acid RMSE 0.247 Creatinine R^2 0.944 Creatinine RMSE 0.116	uric acid R^2 0,974 uric acid RMSE 0.314 Creatinine R^2 0.941 Creatinine RMSE 0.168 Hypoxanthine R^2 0.974 Hypoxanthine RMSE 0.089	uric acid R^2 0.976 uric acid RMSE 0.255 Creatinine R^2 0.940 Creatinine RMSE 0.143 Hypoxanthine R^2 0.977 Hypoxanthine RMSE 0.099 Xanthine R^2 0.971 Xanthine RMSE 0.095

solution, as well as spectra of the mixed solution, as depicted in Fig. 3. By aligning the mixed solution spectra with the pure spectra of each metabolite, it was observed that the spectral information of uric acid, hypoxanthine, creatinine, xanthine, and ergothioneine could be partially discerned in the mixed solution spectra. However, some spectral features were overlapped or disappeared. The characteristic peaks of creatinine at 869 and 930 cm^{-1} were absent in the mixed spectra, while the peaks at 1426 and 617 cm^{-1} overlapped with those of other substances at nearby wavenumbers, becoming more evident only at higher concentrations with stronger relative peak intensities. Hypoxanthine's SERS characteristic peak at 725 cm^{-1} was discernible, but other peaks were obscured by the peaks of ergothioneine and other substances. Xanthine's SERS peaks at 654 and 957 cm^{-1} were more evident in the mixed spectra, with other peaks only slightly discernible at higher concentrations, such as those at 1245 and 1320 cm^{-1}. The characteristic peak of uric acid at 635 cm^{-1} was almost completely overshadowed by the peak of xanthine at 654 cm^{-1}, while the peak at 810 cm^{-1} was more distinct in the mixed spectra, with other peaks either disappearing or highly overlapping with those of other substances. Ergothioneine, as an interfering factor, was particularly evident in the mixed spectra, especially its five broad SERS peaks in the 1097–1527 cm^{-1} range, which obscured a significant number of SERS characteristic peaks of the four target substances within that wavenumber range, posing difficulties in spectral information extraction.

3.3. Performance evaluation of multiple metabolites quantitative detection model

In this section, we validated the model's quantitative performance

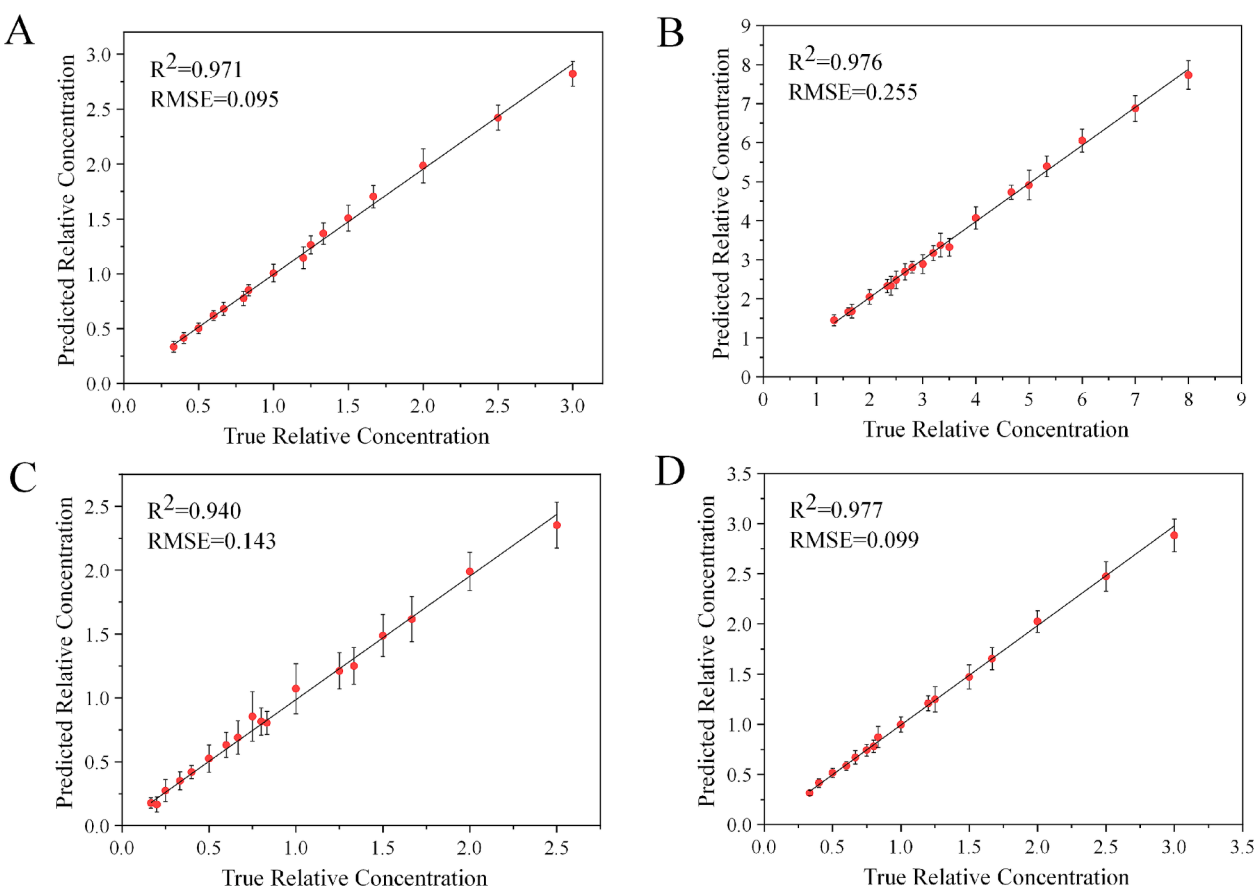

Fig. 4. Concentration prediction performance of the model for four metabolites. (A) Xanthine, (B) Uric acid, (C) Creatinine, (D) Hypoxanthine. (A–C) X-axis: Ture relative concentration of xanthine, uric acid, and creatinine, respectively, relative to hypoxanthine. Y-axis: Predicted relative concentration by the model. (D) X-axis: True relative concentration of hypoxanthine relative to xanthine. Y-axis: Predicted relative concentration of hypoxanthine relative to xanthine by the model.

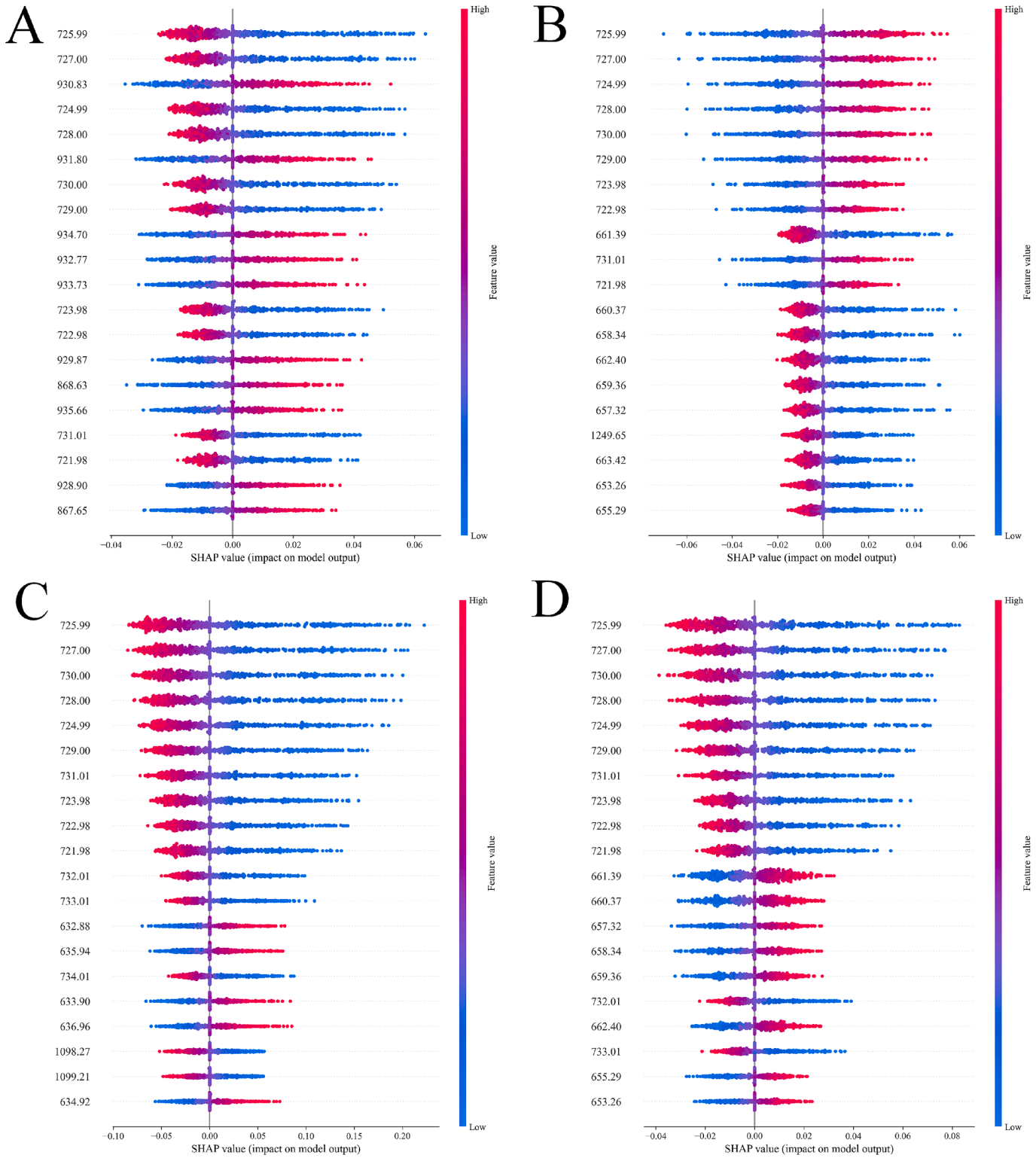

Fig. 5. SHAP feature density scatter plots explaining the quantification effects of metabolites (A) Creatinine (B) Hypoxanthine(C) Uric Acid(D) Xanthine.

and examined if the simultaneous quantification of multiple metabolites could diminish the predictive accuracy. A dataset of 540 spectra was randomly shuffled in a 7:2:1 ratio to form training, validation, and test sets, ensuring the independent distribution of the test set. The training and validation sets were used for model training and tuning, with the model's performance evaluated in the test set.

Initially, we employed the neural network architecture to independently model and predict the concentration of each target metabolite. The determination coefficients (R^2) for the predicted concentrations versus actual concentrations for the four models are shown in Table 1.

The model method demonstrated satisfactory performance in the single-substance quantitative detection of uric acid, hypoxanthine, and xanthine across the 540 spectra obtained. Creatinine exhibited an R^2 value that was 0.03 lower than the other metabolites due to its spectral information being significantly masked in the mixed spectra. Nonetheless, the value is still deemed acceptable and has the potential for

enhancement through the enrichment of the training data. This substantiated the model's performance in quantitative regression.

Subsequently, we incrementally increased the number of metabolites quantified simultaneously by the model. The quantitative results when using the neural network architecture model to simultaneously quantify two, three, and four metabolite targets are shown in Table 2.

Table 2 indicates that the quantitative performance for each metabolite did not significantly decrease as the number of metabolites quantified simultaneously increased. This demonstrates the multi-channel neural network architecture's potential for effective quantification of a greater number of target substances.

To further verify the model's stability, in the assessment of simultaneous quantification of four metabolites, this study constructed 10 batches of training and testing data using a 10-fold cross-validation method. The average results of the 10 training and validation measurements were used to estimate the model performance under the neural network architecture. The quantification effects for each metabolite in each of the 10 runs are detailed in the Supplementary Material (Table S1), with the average results illustrated in the Fig. 4.

In our study, we employed the SHAP method to provide a visual explanation of the prediction process within our multi-channel feature extraction neural network model. Fig. 5 illustrates the feature density scatter plots obtained through the SHAP method, which elucidates the magnitude of influence each characteristic peak exerts on the predictive outcome. The top 20 most influential wavenumbers are listed. These wavenumbers are then mapped to the regions around specific characteristic peaks, allowing us to identify the most significant feature peaks. Each dot represents a sample, with redder colors indicating higher feature values and bluer colors indicating lower feature values. The further a dot is from the central line (zero point), the greater the impact of the feature on the model's output. Positive SHAP values signify a positive impact, while negative SHAP values indicate a negative impact.

Fig. 5A, C, and D correspond to the quantification of Creatinine, Uric Acid, and Xanthine, respectively, with Hypoxanthine serving as the internal standard. From Fig. 5A, C, and D, it is evident that the feature peak near 725 cm^{-1} exerts a negative impact on quantification. The 725 cm^{-1} peak is a notably prominent feature peak for Hypoxanthine, and the stronger the relative intensity of the internal standard peak, the lower the corresponding quantified substance concentration.

In Fig. 5A, the feature peak near 930, 869 cm^{-1} is observed to positively influence the quantification of Creatinine, suggesting that the stronger the intensity of the feature peak in this region, the higher the Creatinine concentration. Upon comparison with the reference spectrum, the peaks at 930, 869 cm^{-1} indeed correspond to two characteristic peaks of Creatinine. In Fig. 5C, the feature peak that positively affects the quantification of Uric Acid is located near 632 cm^{-1}, which corresponds to a characteristic peak of Uric Acid. Similarly, in Fig. 5D, the feature peak that positively influences the quantification of Xanthine is near 660 cm^{-1}, aligning with the characteristic peak of Xanthine. Fig. 5B represents the quantification of Hypoxanthine using Xanthine as the internal standard. It can be observed that the feature peak corresponding to Hypoxanthine near 725 cm^{-1} positively influences the quantification of Hypoxanthine, while the feature peak of the internal standard Xanthine near 660 cm^{-1} negatively impacts the quantification of Hypoxanthine.

The incorporation of SHAP-guided interpretation within our multi-channel feature extraction neural network has yielded a deeper understanding of the model's decision-making process. It is apparent that the model accurately identified the respective quantitative characteristic peaks for each substance in the mixture spectrum and achieved a commendable simultaneous quantification effect based on these features. This insight not only strengthens the trust in our model's predictions but also opens avenues for the discovery of novel biomarkers and the elucidation of metabolic pathways.

Finally, a comparative test was conducted between this method and the PLS-regression method for the simultaneous quantification of multiple metabolites, with the results shown in Table 3.

Table 3
Comparison of multi-target quantitative prediction performance between multi-channel neural network models and PLS-Regression.

Model		Xanthine	Hypoxanthine	Uric Acid	Creatinine
Multi-Channel Neural Network	R^2	0.971	0.977	0.976	0.940
	RMSE	0.095	0.099	0.255	0.143
PLS Regression	R^2	0.823	0.902	0.855	0.813
	RMSE	0.272	0.232	0.741	0.223

Table 3 demonstrates that the neural network model's concentration prediction performance for each metabolite significantly outperformed that of the PLS-Regression model, showcasing the model's capability for simultaneous multi-target substance quantification.

4. Conclusion

Our study successfully integrates label-free Surface-Enhanced Raman Spectroscopy (SERS) with a deep learning algorithm to achieve the simultaneous quantitative detection of multiple metabolites. This innovative approach has been validated through the accurate prediction of xanthine, hypoxanthine, uric acid, and creatinine, with determination coefficients (R^2) of 0.971, 0.977, 0.976, and 0.940, respectively. The integration of SHAP-guided interpretation within our deep learning framework has provided a deeper understanding of the model's decision-making process, revealing the specific spectral peaks that contribute to the quantification of the target metabolites. This insight not only strengthens the trust in our model's predictions but also opens avenues for the discovery of novel biomarkers and the elucidation of metabolic pathways. This method offers a promising alternative to traditional analytical techniques, providing a rapid, sensitive, and cost-effective means for metabolite analysis with potential clinical applications. The results underscore the potential of this technology to transform early disease diagnosis and contribute to personalized medicine. Future work will focus on expanding the applicability of this method to a broader range of clinical samples and further refining the model for enhanced predictive capabilities.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Moonshot AI in order to improve language and readability, with caution. After using this tool, the author reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.

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Data availability

Data will be made available on request.

References

[1] A. Zhang, H. Sun, G. Yan, et al., Metabolomics for biomarker discovery: moving to the clinic, Biomed Res. Int. (2015, 2015.).
[2] F.R. Pinu, S.A. Goldansaz, J. Jaine, Translational metabolomics: current challenges and future opportunities, Metabolites 9 (6) (2019) 108.
[3] K. Ortmayr, S. Dubuis, M. Zampieri, Metabolic profiling of cancer cells reveals genome-wide crosstalk between transcriptional regulators and metabolism, Nat. Commun. 10 (1) (2019) 1841.
[4] B. Yuan, S. Schafferer, Q. Tang, et al., A plasma metabolite panel as biomarkers for early primary breast cancer detection, Int. J. Cancer 144 (11) (2019) 2833–2842.
[5] M. Kdadra, S. Höckner, H. Leung, et al., Metabolomics biomarkers of prostate cancer: a systematic review, Diagnostics 9 (1) (2019) 21.
[6] C. Qin, G. Yang, J. Yang, et al., Metabolism of pancreatic cancer: paving the way to better anticancer strategies, Mol. Cancer 19 (2020) 1–19.
[7] S. Bamji-Stocke, V. van Berkel, D.M. Miller, et al., A review of metabolism-associated biomarkers in lung cancer diagnosis and treatment, Metabolomics 14 (2018) 1–16.
[8] G.J. Patti, O. Yanes, G. Siuzdak, Metabolomics: the apogee of the omics trilogy, Nat. Rev. Mol. Cell Biol. 13 (4) (2012) 263–269.
[9] N. Feizi, F.S. Hashemi-Nasab, F. Golpelichi, et al., Recent trends in application of chemometric methods for GC-MS and GC×GC-MS-based metabolomic studies, TrAC Trends Anal. Chem. 138 (2021) 116239.
[10] A. Zhang, H. Sun, P. Wang, et al., Modern analytical techniques in metabolomics analysis, Analyst 137 (2) (2012) 293–300.
[11] Z. Li, M. Wang, Y. Jiao, et al., Different number of silver nanoparticles layers for surface enhanced Raman spectroscopy analysis, Sens. Actuators B 255 (2018) 374–383.
[12] S. Xu, S. Jiang, J. Wang, et al., Graphene isolated Au nanoparticle arrays with high reproducibility for high-performance surface-enhanced Raman scattering, Sens. Actuators B 222 (2016) 1175–1183.
[13] N. Kim, M.R. Thomas, M.S. Bergholt, et al., Surface enhanced Raman scattering artificial nose for high dimensionality fingerprinting, Nat. Commun. 11 (1) (2020) 207.
[14] Y. Liu, M. Li, H. Liu, et al., Cancer diagnosis using label-free SERS-based exosome analysis, Theranostics 14 (5) (2024) 1966.
[15] W. Zhu, J.A. Hutchison, M. Dong, et al., Frequency shift surface-enhanced Raman spectroscopy sensing: An ultrasensitive multiplex assay for biomarkers in human health, ACS Sens. 6 (5) (2021) 1704–1716.
[16] J. Chen, J. Wang, Y. Geng, et al., Single-cell oxidative stress events revealed by a renewable SERS nanotip, ACS Sensors 6 (4) (2021) 1663–1670.
[17] Y. Lu, L. Lin, J. Ye, Human metabolite detection by surface-enhanced Raman spectroscopy, Mater. Today Bio 13 (2022) 100205.
[18] E. Avci, H. Yilmaz, N. Sahiner, et al., Label-free surface enhanced Raman spectroscopy for cancer detection, Cancers 14 (20) (2022) 5021.
[19] H. Shin, B.H. Choi, O. Shim, et al., Single test-based diagnosis of multiple cancer types using Exosome-SERS-AI for early stage cancers, Nat. Commun. 14 (1) (2023) 1644.
[20] H. Ge, X. Gao, J. Lin, et al., Label-free SERS detection of prostate cancer based on multi-layer perceptron surrogate model method, Spectrochim. Acta A Mol. Biomol. Spectrosc. 304 (2024) 123407.
[21] X. Bai, J. Lin, X. Wu, et al., Label-free detection of bladder cancer and kidney cancer plasma based on SERS and multivariate statistical algorithm, Spectrochim. Acta A Mol. Biomol. Spectrosc. 279 (2022) 121336.
[22] X. Su, X. Liu, Y. Xie, et al., Quantitative label-free SERS detection of trace fentanyl in biofluids with a freestanding hydrophobic plasmonic paper biosensor, Anal. Chem. 95 (7) (2023) 3821–3829.
[23] M. Zareef, M.M. Hassan, M. Arslan, et al., Rapid prediction of caffeine in tea based on surface-enhanced Raman spectroscopy coupled multivariate calibration, Microchem. J. 159 (2020) 105431.
[24] Y. Zhang, L. Jiang, Y. Chen, et al., Detection of chlorpyrifos residue in apple and rice samples based on aptamer sensor: Improving quantitative accuracy with partial least squares model, Microchem. J. 194 (2023) 109352.
[25] G. Fang, X. Lin, X. Liang, et al., Machine learning-driven 3D plasmonic cavity-in-cavity surface-enhanced raman scattering platform with triple synergistic enhancement toward label-free detection of antibiotics in milk, Small 18 (45) (2022) 2204588.
[26] Y. Wen, X. Wang, D. Li, et al., Rapid detection of phenytoin sodium by partial-least squares and linear regression models combined with surface-enhanced Raman spectroscopy, J. Pharm. Biomed. Anal. 223 (2023) 115160.
[27] D. Li, Q. Zhang, B. Deng, et al., Rapid, sensitive detection of ganciclovir, penciclovir and valacyclovir-hydrochloride by artificial neural network and partial least squares combined with surface enhanced Raman spectroscopy, Appl. Surf. Sci. 539 (2021) 148224.
[28] S. Lin, X. Fang, G. Fang, et al., Ultrasensitive detection and distinction of pollutants based on SERS assisted by machine learning algorithms, Sens. Actuators B 384 (2023) 133651.
[29] J.B. Phyo, A. Woo, H.J. Yu, et al., Label-free SERS analysis of urine using a 3D-stacked AgNW-glass fiber filter sensor for the diagnosis of pancreatic cancer and prostate cancer, Anal. Chem. 93 (8) (2021) 3778–3785.
[30] Q. Yuan, L.F. Yao, J.W. Tang, et al., Rapid discrimination and ratio quantification of mixed antibiotics in aqueous solution through integrative analysis of SERS spectra via CNN combined with NN-EN model, J. Adv. Res. (2024).
[31] F. Zhao, W. Wang, H. Zhong, et al., Robust quantitative SERS analysis with relative Raman scattering intensities, Talanta 221 (2021) 121465.
[32] S.Y. Long, Z.P. Chen, Y. Chen, et al., Quantitative detection of captopril in tablet and blood plasma samples by the combination of surface-enhanced Raman spectroscopy with multiplicative effects model, J. Raman Spectrosc. 46 (7) (2015) 605–609.
[33] G. Fang, W. Hasi, X. Lin, et al., Automated Identification of Pesticide Mixtures via Machine Learning Analysis of TLC-SERS Spectra, J. Hazard. Mater. (2024) 134814.
[34] X. Lin, G. Fang, Y. Liu, et al., Marangoni effect-driven transfer and compression at three-phase interfaces for highly reproducible nanoparticle monolayers, J. Phys. Chem. Lett. 11 (9) (2020) 3573–3581.
[35] S.M. Lundberg, S.I. Lee, A unified approach to interpreting model predictions, Adv. Neural Inf. Process. Syst. 30 (2017).